

$$\rho = 3850 \text{ kg/m}^3$$

$$D = 6480 \text{ km} = 6480000 \text{ m}$$

$$r = \frac{D}{2} = 3240000 \text{ m}$$

$$V = \frac{4}{3} r^3 \pi = 1.638 \cdot 10^{20} \text{ m}^3$$

$$M = \rho \cdot V = 6.48 \cdot 10^{23}$$

$$\alpha_M = G \cdot \frac{M}{r^2}$$

$$= 6.67 \cdot 10^{-11} \cdot \frac{6.48 \cdot 10^{23}}{3240000^2} = 3.74 \text{ m/s}^2$$

$$\frac{g_M}{g_E} = 0.38 \rightarrow \text{razlika}$$

$$v = \sqrt{2gh} = \sqrt{2 \cdot 3.74 \cdot 10} = 8.65 \text{ m/s}$$

$$\textcircled{2} \text{ a) } m = 2 \text{ kg} \\ x = 3.5 \text{ m} \\ W = 30.5 \text{ J}$$

$$\text{b) } v = 4 \text{ m/s} \\ E_k = \frac{mv^2}{2} = \frac{2 \cdot 4^2}{2} = 16 \text{ J}$$

$$E_{\text{tot}} = W + E_k = 46.5 \text{ J}$$

$$\textcircled{3} \text{ } f_r = 30 \text{ FPS} \quad \text{a) } a = \frac{2 \cdot \Delta x}{(\Delta t)^2} = \frac{2 \cdot 1.353}{\left(\frac{30}{650}\right)^2} = 12.7 \text{ m/s}^2 \\ f_s = 650$$

$$\text{b) } m = 2.45$$

$$F = m \cdot a = 31.12 \text{ N}$$

$$\text{c) } I = F \cdot \Delta t = 31.12 \cdot 0.04 = 26.97 \text{ Ns}$$

$$\text{d) } v_1 = \frac{\Delta x_1}{\Delta t_1} = 18.08 \text{ m/s} \quad v_2 = \frac{\Delta x_2}{\Delta t_2} = 21.09 \text{ m/s}$$

$$\textcircled{4} \text{ } \rho_H = 3950 \text{ kg/m}^3 \quad v_0 = 0 \text{ m/s} \\ r = 6790 \text{ km} = 6790000 \text{ m} \\ h = 10 \text{ m}$$

$$g = G \cdot \frac{M}{r^2} = 6.67 \cdot 10^{-11} \cdot \frac{6 \cdot 10^{24}}{(6400 \cdot 1000)^2} \\ = 9.77 \text{ m/s}^2$$

$$V_H = \frac{4}{3} r^3 \pi = 1.31 \cdot 10^{21}$$

$$\rho_H = \frac{M_H}{V_H}$$

$$M_H = \frac{V}{\rho} = \frac{1.31 \cdot 10^{21}}{3950} = 3.32 \cdot 10^{17}$$

$$g_H = G \cdot \frac{M}{r^2} =$$

$$M_H = 6.81 \cdot 10^{23}$$

1) $m = 10 \text{ kg}$

$$T = F_g \cdot \sin(\alpha)$$

$$\alpha = 30^\circ$$

$$T = 98.1 \text{ N} \cdot \sin(30) = 49.05 \text{ N}$$

$$F_g = m \cdot g = 10 \cdot 9.81 = 98.1 \text{ N}$$



2) $v_0 = 2 \text{ m/s}$

$$m = 0.55 \text{ kg}$$

$$g = 9.81 \text{ m/s}^2$$

$$\Delta t = 0.1 \text{ s}$$

$$h = 1.5 \text{ m}$$

$$k = 0.25 \text{ kg/m}$$

n	t	x	v	a
0	0	1.5	2	-11.63
1	0.1	1.58	0.84	-10.13
2	0.2	1.56	-0.14	-9.80

$$a_0 = \frac{-k v_0^2}{m} - g = -11.63 \text{ m/s}^2$$

$$v_1 = v_0 + a_0 \cdot \Delta t = 0.84 \text{ m/s}$$

$$x_1 = x_0 + v_1 \cdot \Delta t = 1.58 \text{ m}$$

$$a_1 = \frac{-k v_1^2}{m} - g = -10.13 \text{ m/s}^2$$

$$v_2 = v_1 + a_1 \cdot \Delta t = -0.143 \text{ m/s}$$

$$x_2 = x_1 + v_2 \cdot \Delta t = 1.56 \text{ m}$$

$$a_2 = \frac{-k v_2^2}{m} - g = -9.80 \text{ m/s}^2$$

3) $m_L = 300 \text{ kg}$

$$F_L = m \cdot a_T = 300 \cdot 1.36 = 408 \text{ N}$$

$$g_E = G \cdot \frac{M_E}{r^2} = 6.67 \cdot 10^{-11} \cdot \frac{6 \cdot 10^{24}}{(6378.1 \cdot 10^3)^2} = 9.84 \text{ m/s}^2$$

$$g_T = G \cdot \frac{M_T}{r_T^2} = 6.67 \cdot 10^{-11} \cdot \frac{1.35 \cdot 10^{22}}{(2575 \cdot 10^3)^2} = 1.36 \text{ m/s}^2$$

$$\frac{g_T}{g_E} = 0.14$$

4) $W_i = 3.1 \text{ eV}$

$$f = ?$$

$$\lambda = ?$$

$$p = \frac{W_i}{h}$$

$$= \frac{3.1 \text{ eV}}{6.626 \cdot 10^{-34}}$$

$$f = 7.49 \cdot 10^{14} \text{ Hz}$$

$$\lambda = \frac{c}{f}$$

$$= \frac{3 \cdot 10^8}{7.49 \cdot 10^{14}} = 4.01 \cdot 10^{-7} \text{ m} = 400 \text{ nm}$$

6) a) $A = 2.24 \text{ mm/s}^2 = 2.24 \cdot 10^{-3} \text{ m/s}^2$

$$a = \frac{2 \cdot A}{f_k} = \frac{2 \cdot 2.24 \cdot 10^{-3}}{\left(\frac{30}{650}\right)} = 2.1 \text{ m/s}^2$$

b) $v = \frac{\Delta x}{\Delta t} = \frac{1.98 - 1.24}{0.1} = 0.052 \text{ m/s}$

c) $v = \frac{n \cdot \lambda}{f_k} = \frac{0.184 \cdot 10^{-1}}{\left(\frac{30}{650}\right)} = 0.4 \text{ m/s}$

d) $m_1 v_1 + m_2 v_2 = m_1 v_1' + m_2 v_2'$

$$m_1 \cdot 0.052 + 100 \cdot 0 = m_1 \cdot 0 + 0.1 \cdot 0.4$$

$$m_1 = 0.74 \text{ kg}$$

$$\begin{aligned}
 \textcircled{1} \quad m &= 4t = 4000 \text{ kg} & F &= m \cdot g \\
 h_0 &= 0 \text{ m} & &= 4000 \cdot 9.81 \\
 \alpha &= 15^\circ & &= 39240 \text{ N} \\
 \mu &= 0.1 \\
 v &= 90 \text{ km/h}
 \end{aligned}$$

$\mu \rightarrow$ koeficient tregja

$$\begin{aligned}
 F_{\text{tr}} &= \mu N = \mu \cdot m \cos \alpha \\
 &= 0.1 \cdot 4000 \cdot \cos 15^\circ = 3810.23 \text{ N} \\
 &= 3830.23 \text{ N}
 \end{aligned}$$

$$F_{\text{uk}} = mg \sin \theta - \mu mg \cos \theta$$

$$\begin{aligned}
 F_{\text{uk}} &= mg(\sin \alpha - \mu \cos \alpha) \\
 &= 4000 \cdot 9.81 \cdot (\sin 15^\circ - 0.1 \cdot \cos 15^\circ) \\
 &= 6365.77 \text{ N}
 \end{aligned}$$

$$\begin{aligned}
 a &= g(\sin \alpha - \mu \cos \alpha) \\
 &= 9.81(\sin 15^\circ - 0.1 \cos 15^\circ) \\
 &= 1.59 \text{ m/s}^2
 \end{aligned}$$

$$v^2 = v_0^2 + 2as$$

$$0 = v_0^2 + 2as$$

$$s = \frac{v_0^2}{2a}$$

$$s = \frac{25^2}{2 \cdot 1.59} = 154.17 \text{ m} = 136.54 \text{ m}$$

$$\begin{aligned}
 \textcircled{2} \quad W_i &= 2.5 \text{ eV} = 4 \cdot 10^{-18} \text{ J} \\
 E_k &= 1 \text{ eV} = 1.6 \cdot 10^{-19} \text{ J} \\
 f &= f + 1 \cdot 10^{15} \text{ Hz} \\
 v &=?
 \end{aligned}$$

$$\begin{aligned}
 f &= \frac{W_i}{h} \\
 f &= \frac{2.5 \cdot 1.6 \cdot 10^{-18}}{6.626 \cdot 10^{-34}} \\
 f_1 &= 6.04 \cdot 10^{14} \\
 f_n &= 6 \cdot 10^{15}
 \end{aligned}$$

$$\begin{aligned}
 f &= \frac{W_i}{2\pi} \\
 &= \frac{2.5 \cdot 1.6 \cdot 10^{-18}}{2\pi} \\
 f_0 &= 6.34 \cdot 10^{-20} \\
 f_n &=
 \end{aligned}$$

$$h \cdot f = W_i + E_k$$

$$f = \frac{W_i + E_k}{h}$$

$$f_0 = 7.55 \cdot 10^{14} \text{ Hz}$$

$$f_n = 1.75 \cdot 10^{15} \text{ Hz}$$

$$E_{\text{total}} = W_i + E_k = 2.5 + 1 = 3.5 \text{ eV}$$

② $v_0 = 2 \text{ m/s}$
 $m = 0.55 \text{ kg}$
 $g = -9.81 \text{ m/s}^2$
 $\Delta t = 0.1 \text{ s}$
 $x = 1.5 \text{ m}$

n-korak	trajektorija	položaj	brzina	ubrzanje
0	0 s	1.5 m	2 m/s	-11.63 m/s ²
1	0.1 s	1.58 m	0.84 m/s	-10.03 m/s ²
2	0.2 s	1.56 m	-0.18 m/s	-9.83 m/s ²
3	0.3 s	1.44 m	-1.16 m/s	-9.19 m/s ²
4	0.4 s	1.23 m	-2.08 m/s	-7.84 m/s ²

$$a_0 = \frac{-kv^2}{m} - g = \frac{-0.25 \cdot 2^2}{0.55} - 9.81 = -11.63 \text{ m/s}^2$$

$$v_1 = v_0 + a_0 \cdot \Delta t = 2 - 11.63 \cdot 0.1 = 0.84 \text{ m/s}$$

$$x_1 = x_0 + v_1 \cdot \Delta t = 1.5 + 0.84 \cdot 0.1 = 1.58 \text{ m}$$

$$a_1 = \frac{-kv^2}{m} - g = \frac{-0.25 \cdot 0.84^2}{0.55} - 9.81 = -10.13 \text{ m/s}^2$$

$$v_2 = v_1 + a_1 \cdot \Delta t = 0.84 - 10.13 \cdot 0.1 = -0.183 \text{ m/s}$$

$$x_2 = x_1 + v_2 \cdot \Delta t = 1.58 - 0.183 \cdot 0.1 = 1.56 \text{ m}$$

$$a_2 = \frac{-kv^2}{m} - g = \frac{-0.25 \cdot (-0.183)^2}{0.55} - 9.81 = -9.83 \text{ m/s}^2$$

$$v_3 = -0.18 - 9.83 \cdot 0.1 = -1.16 \text{ m/s}$$

$$x_3 = 1.56 - 1.16 \cdot 0.1 = 1.44 \text{ m}$$

$$a_3 = \frac{-kv^2}{m} - g = \frac{-0.25 \cdot (-1.16)^2}{0.55} - 9.81 = -9.19 \text{ m/s}^2$$

$$v_4 = -1.16 - 9.19 \cdot 0.1 = -2.08 \text{ m/s}$$

$$x_4 = 1.44 - 2.08 \cdot 0.1 = 1.23 \text{ m}$$

$$a_4 = \frac{-kv^2}{m} - g = \frac{-0.25 \cdot (-2.08)^2}{0.55} - 9.81 = -7.84 \text{ m/s}^2$$

~~XXXXXXXXXX~~

Fotoefekt

① Najmanjša frek. svetlosti potrebna za fotoelektroni efekt. Izloži tudi je $W_i = 3.1 \text{ eV}$ kolika je valovna dolžina te svetlosti?

$$h = 6.626 \cdot 10^{-34} \text{ Js}$$

$$c = 3 \cdot 10^8$$

$$1 \text{ eV} = 1.6 \cdot 10^{-19} \text{ J}$$

$$W_i = 3.1 \text{ eV}$$

$$W_i = h \nu_{\min}$$

$$\nu_{\min} = \frac{W_i}{h}$$

$$\nu = \frac{3.1 \cdot 10^{-19} \text{ J}}{6.626 \cdot 10^{-34} \text{ Js}} = \frac{4.96 \cdot 10^{-13} \text{ s}^{-1}}{6.626 \cdot 10^{-34} \text{ Js}} = 7.485662542 \cdot 10^{14} \text{ s}^{-1}$$

$$\nu = 7.485662542 \cdot 10^{14} \text{ s}^{-1}$$

$$\lambda = \frac{c}{\nu} = \frac{3 \cdot 10^8}{7.485662542 \cdot 10^{14}} = 4.00766129 \cdot 10^{-7}$$

$$\begin{aligned} \textcircled{5} \quad U_0 &= 6 \text{ eV} \\ L &= ? \text{ nm} \\ E_0 &= 5.2 \text{ eV} \\ T &= 0.002 \end{aligned}$$

$$\textcircled{8} \quad \lambda = 293 \text{ nm}$$

$$E_k = 1.6 \text{ eV}$$

$$\nu_{\max} = ?$$

$$h = 6.626 \cdot 10^{-34}$$

$$W_i = \frac{hc}{\lambda}$$

$$h_0 = W_i + e \cdot U_k$$

$$W_i = h_0 - e \cdot U_k = \frac{h \cdot c}{\lambda} - 1.6$$

$$h = \frac{6.626 \cdot 10^{-34}}{1.6 \cdot 10^{-15}} = 4.14125 \cdot 10^{-15} \text{ eV}$$

$$W_i = \frac{4.14125 \cdot 10^{-15} \cdot 3 \cdot 10^8}{2.93 \cdot 10^{-7}} - 1.6 = 4.240184713 \text{ eV} - 1.6 = 2.6 \text{ eV}$$

$$-2kL = \ln T$$

$$L = \frac{\ln T}{-2k}$$

$$L = 6.485335514 \cdot 10^{-10} = 0.68 \text{ nm}$$

$$k = \sqrt{\frac{8\pi^2 m (U_0 - E_0)}{h^2}} = \sqrt{\frac{8\pi^2 \cdot 9.11 \cdot 10^{-31} \cdot (6 \text{ eV} - 5.2 \text{ eV})}{(6.626 \cdot 10^{-34})^2}} = 4.5739410^9$$

$$(8) L = 0.5 \mu\text{m}$$

$$T = 0.005$$

$$U = 2.4 \text{ eV} - 0.16 \text{ eV} = 2.24 \text{ eV}$$

$$E_k = 1.6 / 10 = 0.16 \text{ eV}$$

$$L_2 =$$

$$-2\kappa L_2 = \ln(T)$$

$$\kappa = \sqrt{\frac{8\pi^2 m(U - E)}{h^2}} =$$

$$= \sqrt{7384044103} = 7.38 \text{ nm}^{-1}$$

$$-2\kappa L_2 = \ln T$$

$$L_2 = \frac{\ln T}{-2\kappa} = 0.35 \text{ nm},$$

$$(8^*) U_0 = 6 \text{ eV}$$

$$L = 0.7 \text{ nm}$$

$$T = 0.004$$

~~Handwritten scribbles~~

$$4C + 81 + (5 - 1) = (8) \times$$

$$(8a) \lambda = 335 \text{ nm}$$

$$W_i = 1.4 \text{ eV}$$

$$V = ?$$

$$W_i = \frac{h}{\lambda}$$

$$V = \frac{W_i}{h} = \frac{1.4 \text{ eV}}{6.626 \cdot 10^{-34}} = 2.112888621 \cdot 10^{33}$$

b)

$$L = 1 \text{ nm}$$

$$T = 0.02$$

$$E_i = 1.4 - 0.28 = 1.12 \text{ eV}$$

$$U_0 = 0.4 \text{ eV}$$

$$k = \sqrt{\frac{8\pi^2 m (U_0 - E_i)}{h^2}} =$$

$$\sqrt{\frac{78.96 \cdot 9.11 \cdot 10^{-31}}{h^2}}$$

$$k = \sqrt{\frac{3.07116752 \cdot 10^{-29}}{h^2}}$$

$$k = 8.290368266 \cdot 10^{18}$$

$$-2kL = (h\lambda)$$

$$L = \frac{(h\lambda)}{-2k} = 2.359378026 \cdot 10^{-18}$$

Faktor reproduktioje

$$\textcircled{6} \quad f_r = 30 \\ f_g = 650$$

$$a) \quad Q = 2A$$

$$A = 2.24 \text{ mm/s}^2 = 0.0224 \text{ m/s}^2$$

$$\alpha = \frac{2 \cdot 0.0224}{\left(\frac{30}{650}\right)^2} = 21.031 \text{ m/s}^2$$

$$b) \quad v_{b_0} = \frac{\Delta x}{\Delta t} = \frac{1.48 - 1.24}{0.01} = 11266.66667 \text{ mm/s} = 11.27 \text{ m/s}$$

$$c) \quad v = \frac{0.184 \cdot 10^{-1} \text{ m/s}}{K} = \frac{0.184 \cdot 10^{-1} \text{ m/s}}{\left(\frac{30}{650}\right)^2} = 8.638 \text{ m/s}$$

$$d) \quad m_b = ? \\ v_{b_0} = 11.27 \text{ m/s} \\ m_x = 100 \text{ g} = 0.1 \text{ kg} \\ v_{x_0} = 0 \text{ m/s} \\ v_{x_1} = 8.638 \text{ m/s} \\ v_{b_1} = 0 \text{ m/s}$$

$$m_b \vec{v}_{b_0} + m_x \vec{v}_{x_0} = m_b \vec{v}_{b_1} + m_x \vec{v}_{x_1}$$

$$m_b \cdot 11.27 + 0 \cdot 0 = m_b \cdot 0 + 0.1 \cdot 8.638$$

$$m_b \cdot 11.27 \text{ m/s} = 0.8638 \text{ kg/m/s}$$

$$m_b = 0.08 \text{ kg}$$

$$P1.3) \quad a) \quad \alpha = A = 1.353 \text{ cm/s}^2$$

$$\alpha = \frac{2 \cdot 1.353}{\left(\frac{30}{650}\right)^2} = 12.17 \text{ m/s}^2$$

$$b) \quad m = 2.95 \text{ kg}$$

$$F = m \cdot \alpha = 31.12 \text{ N}$$

$$c) \quad I = F \cdot t$$

$$= \frac{31.12 \cdot 0.08 \text{ s}}{\left(\frac{30}{650}\right)^2}$$

$$t = 0.16 - 0.08 = 0.08 \text{ s}$$

$$F_g \text{ is } \left(\frac{f_g}{f_r}\right)^2 \text{ sumu zu a)}$$

$$\Delta v_{3-5} = \frac{\Delta x}{\Delta t} = \frac{18.814 - 1.053}{0.08} \\ = \frac{17.761}{0.08} = 2109.8 \\ \downarrow_k \\ = 2.11 \text{ m/s}$$

$$\Delta v_1 = \frac{66.608 - 63.27}{0.04} = 1808.08 \text{ cm/s} = 18.08 \text{ m/s}$$

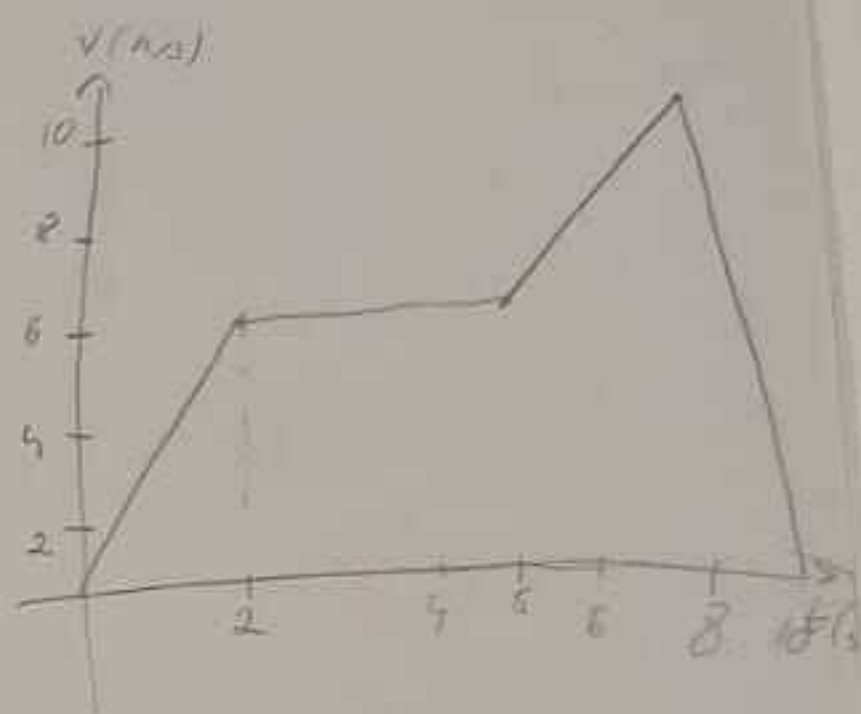
$$\Delta v_2 = 21.038 \text{ m/s}$$

FR 8.8.2024

Kinematika

82) v, t graf; brzina v , vrijeme t , os x

- koliko put je tijelo prešlo
- kolika je prosječna brzina tijela $t_1 - t_2$
- kolika je prosječna brzina tijela tijekom 10s
- Nacrtaj graf ovisnosti



a) $s_1 = 6m$ $s_2 = 63m$
 $s_3 = 18m$
 $s_4 = 24m$
 $s_5 = 0m$
 $s_6 = 0m$

b) $\bar{v} = \frac{\Delta s}{\Delta t} = 6.3 m/s$

c) $\bar{v} = 3 m/s$

d) $a = \frac{\Delta v}{\Delta t}$

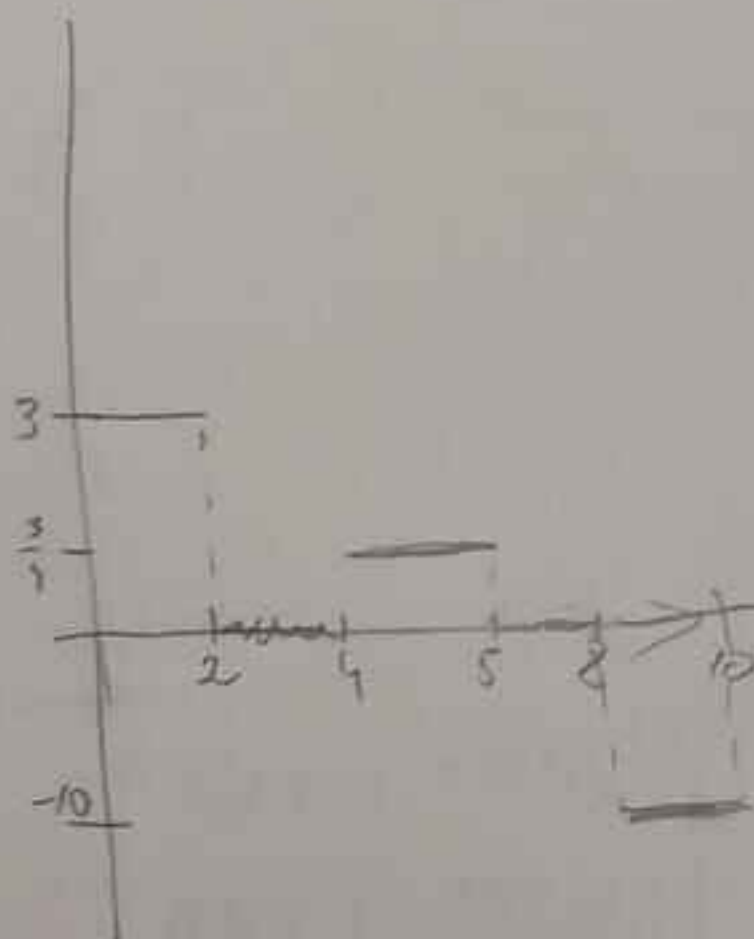
$a_1 = \frac{6-0}{2} = 3 m/s^2$

$a_2 = 0 m/s^2$

$a_3 = \frac{10-6}{3} = \frac{4}{3} m/s^2$

$a_4 = 0$

$a_5 = \frac{0-10}{1} = -10 m/s^2$



• Odredi položaj u 8. sekundi ako je u 2. sekundi -5 m.

$x(8) - x(2) = P_{2-8}$

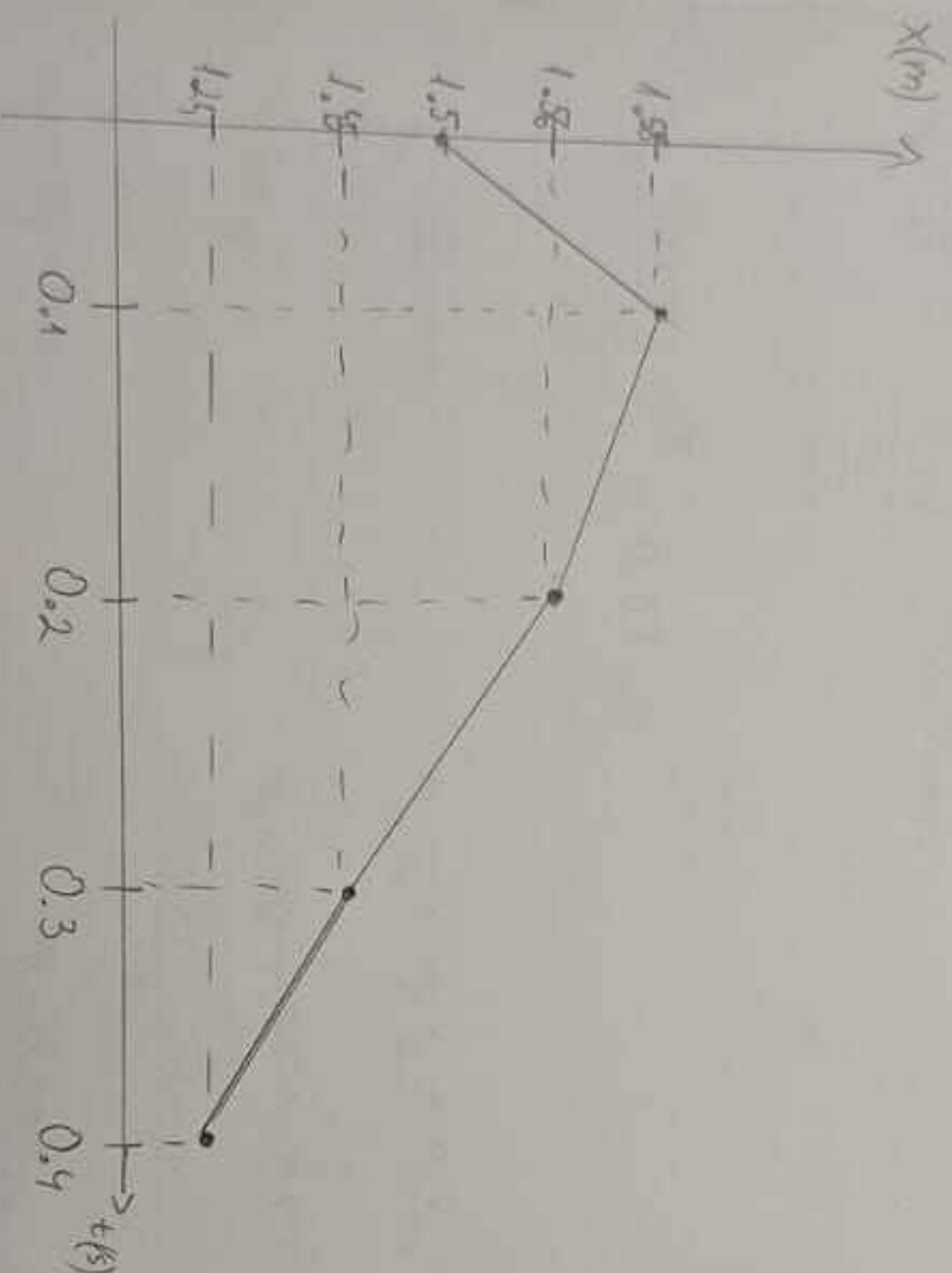
$x(8) = -(-5) + 18 + 24$

$x(2) = 34m$

$x(8) = -(-5) + 18 + 24$

②

n	t	x	v	a
0	0	1.5 m	2 m/s	-11.63 m/s ²
1	0.1	1.58 m	0.84 m/s	-10.13 m/s ²
2	0.2	1.56 m	-0.17 m/s	-9.80 m/s ²
3	0.3	1.45 m	-1.15 m/s	-9.21 m/s ²
4	0.4	1.24 m	-2.07 m/s	-7.86 m/s ²



$$m = 0.55 \text{ kg}$$

$$k = 0.25 \text{ kg/m}$$

$$a_0 = \frac{-k v_0^2}{m} - g = -11.63 \text{ m/s}^2$$

$$v_1 = v_0 + a_0 \cdot \Delta t = 0.84 \text{ m/s}$$

$$x_1 = x_0 + v_1 \cdot \Delta t = 1.58 \text{ m}$$

$$a_1 = \frac{k v_1^2}{m} - g = -10.13 \text{ m/s}^2$$

$$v_2 = v_1 + a_1 \cdot \Delta t = -0.17 \text{ m/s}$$

$$x_2 = x_1 + v_2 \cdot \Delta t = 1.56 \text{ m}$$

$$a_2 = \frac{-k v_2^2}{m} - g = -9.80 \text{ m/s}^2$$

$$v_3 = v_2 + a_2 \cdot \Delta t = -1.15 \text{ m/s}$$

$$x_3 = x_2 + v_3 \cdot \Delta t = 1.45 \text{ m}$$

$$a_3 = \frac{-k v_3^2}{m} - g = -9.21 \text{ m/s}^2$$

$$v_4 = v_3 + a_3 \cdot \Delta t = -2.07 \text{ m/s}$$

$$x_4 = x_3 + v_4 \cdot \Delta t = 1.24 \text{ m}$$

$$a_4 = \frac{-k v_4^2}{m} - g = -7.86 \text{ m/s}^2$$

A: lopta mijenja smjer u intervalu između 1. i 2. koraka tj. $t = 0.1 \text{ s} - t = 0.2 \text{ s}$

④ $W_i = 3.1 \text{ eV}$

$\lambda = ?$

$$W_i = \frac{h}{\lambda}$$



$$\lambda = \frac{W_i}{h} = \frac{3.1 \cdot 1.6 \cdot 10^{-19}}{6.626 \cdot 10^{-34}} = 7.48 \cdot 10^{-14} \text{ m}$$

$$\lambda = \frac{c}{\nu} = \frac{3 \cdot 10^8 \text{ m/s}}{7.48 \cdot 10^{14} \text{ s}^{-1}} = 4.005 \cdot 10^{-7} \text{ m}$$

⑥ $f_r = 30$

$f_s = 650$

a) $A = 2.24 \text{ mm/s}^2$

$a = 2A$

$$a = \frac{2 \cdot A}{\left(\frac{f_r}{f_s}\right)^2} = \frac{4.48}{\left(\frac{30}{650}\right)^2} = 2103.11 \cdot 10^{-3} = 2.103 \text{ m/s}^2$$

b) $v_b = \frac{\Delta x}{\Delta t} = \frac{1.48 - 1.24}{\frac{30}{650}} = 0.52 \text{ m/s}$

c) $v_z = \frac{0.184 \cdot 10^{-7}}{\frac{30}{650}} = 0.39 \text{ m/s}$

d) $v_{b0} = 0.52 \text{ m/s}$

$m_{b0} = ?$

$v_{b1} = 0 \text{ m/s}$

$v_{z0} = 0 \text{ m/s}$

$m_z = 100 \text{ g} = 0.1 \text{ kg}$

$v_{z1} = 0.39 \text{ m/s}$

$$m_{b0} v_{b0} + m_z v_{z0} = m_b v_{b1} + m_z v_{z1}$$

$$m_b \cdot 0.52 + 0.1 \cdot 0 = m_b \cdot 0 + 0.1 \cdot 0.39 \text{ m/s}$$

$$0.52 m_b = 0.039$$

$$m_b = 0.075 \text{ kg}$$

$$\textcircled{1} v_{s0} = 26 \text{ m/s} \quad a = ?$$

$$\Delta t = 4 \text{ s}$$

$$v_{s1} = 18 \text{ m/s}$$

$$a = \frac{v_0 - v_1}{t} = \frac{8}{4} = 2 \text{ m/s}^2$$

$$b) m_s = 67 \text{ kg}$$

$$v_{s0} = 18 \text{ m/s}$$

$$v_{s1} = 5 \text{ m/s}$$

$$m_k = ?$$

$$v_{k0} = 0 \text{ m/s}$$

$$v_{k1} = 5 \text{ m/s}$$

$$m_1 v_{10} + m_2 v_{20} = m_1 v_{11} + m_2 v_{21}$$

$$67 \cdot 18 + 0 = 67 \cdot 5 + 5 m_2$$

$$1206 = 335 + 5 m_2$$

$$871 = 5 m_2$$

$$m_2 = 174.2 \text{ kg}$$

$$\textcircled{2} m = 0.5 \text{ kg}$$

$$f_s = 2600$$

$$f_r = 45$$

$$a) s = \Delta x_{2-5}$$

$$= 88.85 - 76.603$$

$$s = 12.241 \text{ m}$$

$$b) A = 3.095 \cdot 10^{-3} \text{ m/s}^2$$

$$a = \frac{2A}{f_r^2} = 19.105 \text{ m/s}^2 = 19.11 \text{ m/s}^2$$

$$c) v_1 = \frac{\Delta x}{\Delta t} = \frac{76.603 - 73.280}{0.04} = 4623.61 \text{ m/s}$$

$$v_2 = \frac{\Delta x}{\Delta t} = \frac{88.85 - 84.954}{0.04} = 5411.11 \text{ m/s}$$

$$d) E_{k1} = \frac{mv^2}{2} = \frac{0.5 \cdot 4623.61^2}{2} = 5344442.36 \text{ J}$$

$$E_{k2} = \frac{mv^2}{2} = \frac{0.5 \cdot 5411.11^2}{2} = 7320027.86 \text{ J}$$

$$E_{k2} - E_{k1} = 1975585.498 \text{ J}$$

$$\textcircled{3} \quad h = 30 \text{ m}$$

$$v_0 = 10 \text{ m/s}$$

$$T = \sqrt{\frac{2H}{g}} = 2.47 \text{ s}$$

$$D = v_0 \cdot t$$

$$= 10 \cdot 2.47$$

$$= 24.7 \text{ m}$$

$$v = \sqrt{v_0^2 + (gt)^2}$$

$$= \sqrt{100 + (9.81 \cdot 2.47)^2}$$

$$v = 26.21 \text{ m/s}$$

$$\boxed{\begin{array}{l} T = \sqrt{\frac{2H}{g}} \\ v = \sqrt{v_0^2 + (gt)^2} \\ D = v_0 t \end{array}} \quad \checkmark$$

$$\textcircled{4} \quad a_m = \frac{1}{6} \cdot a_z$$

$$g_z = G \frac{M}{r^2}$$

$$= 9.77 \text{ m/s}^2$$

$$m g_H h_m = m g_z h_z$$

$$\frac{1}{6} g_z h_m = g_z h_z$$

$$h_m = \frac{g_z h_z}{\frac{1}{6} g_z}$$

$$h_m = \frac{h_z}{\frac{1}{6}}$$

$$h_m = 6 h_z //$$

$$\boxed{\begin{array}{l} m g_1 h_1 = m g_2 h_2 \\ \downarrow \\ g_1 h_1 = g_2 h_2 \end{array}} \quad \checkmark$$

$$\textcircled{5} \quad I_1 = 10 \text{ mW/cm}^2 = 1 \cdot 10^{-6}$$

$$I_2 = 500 \text{ mW/cm}^2 = 1 \cdot 10^{-4}$$

$$L = \frac{10 \log \frac{I_2}{I_1}}{1} = 10 \log \cdot \frac{1 \cdot 10^{-4}}{1 \cdot 10^{-6}} = 20 \text{ dB}$$

$$\boxed{L = 10 \log \left(\frac{I_2}{I_1} \right)} \quad \checkmark$$

$$\textcircled{a} \quad W_i = 4 \text{ eV} = 6.4 \cdot 10^{-19} \text{ J} \quad \left| \quad \cancel{\lambda = \frac{c}{\nu}} = \frac{3 \cdot 10^8 \text{ m/s}}{6 \cdot 10^{14} \text{ s}^{-1}} = 500 \text{ nm} \right.$$

$$v = 6 \cdot 10^6 \text{ m/s}$$

$$\cancel{6.4 \cdot 10^{-19} \text{ J}} = \cancel{h \cdot f} = \cancel{h \cdot \frac{c}{\lambda}}$$

$$\text{or } E_k = \frac{mv^2}{2} = \frac{9.1 \cdot 10^{-31} \cdot (6 \cdot 10^6)^2}{2} = 1.64 \cdot 10^{-17} \text{ J}$$

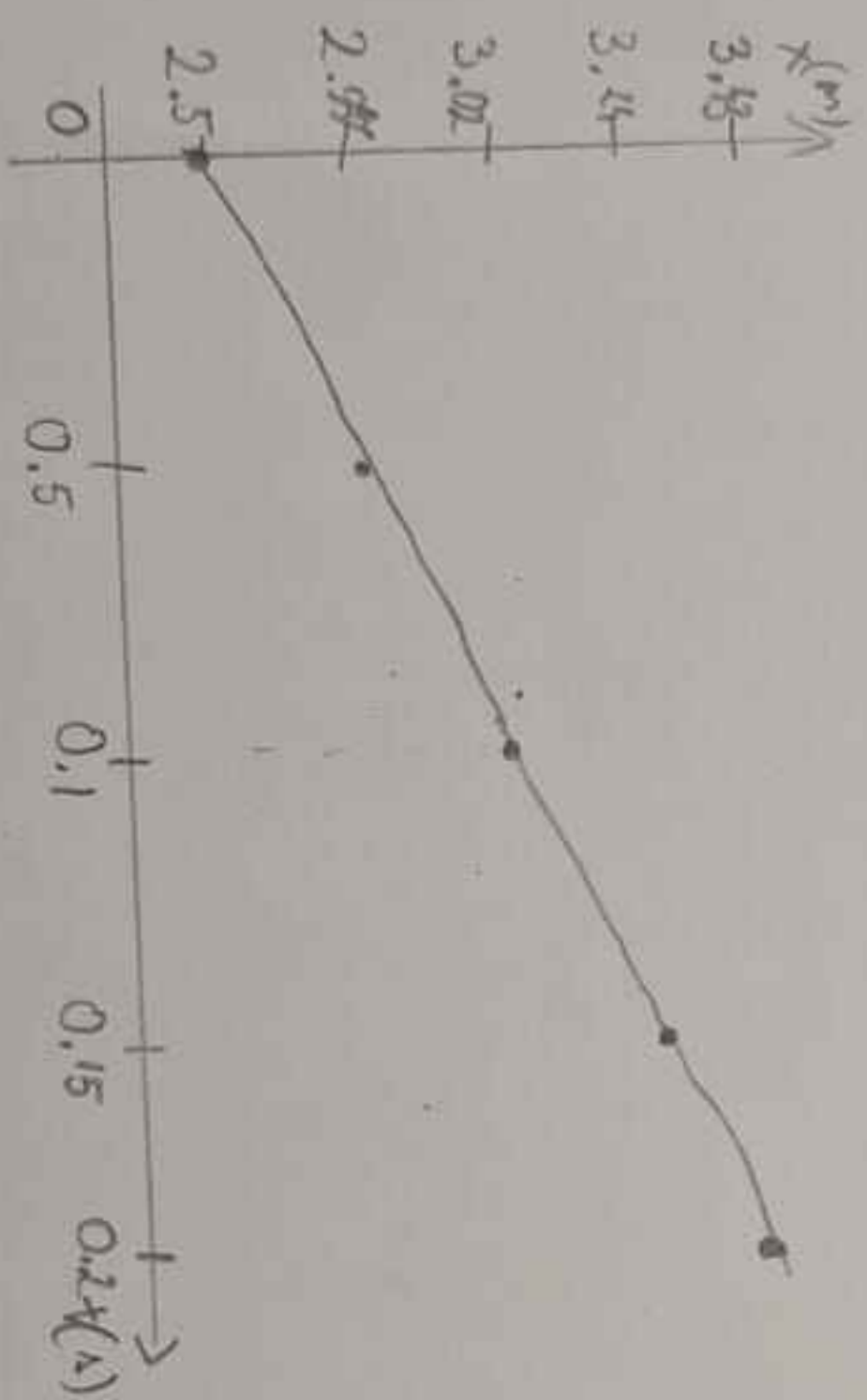
$$h \cdot f = W_i + E_k$$

$$f = \frac{W_i + E_k}{h}$$

$$f = 2.57 \cdot 10^{16} \text{ Hz}$$

$E_k = \frac{mv^2}{2}$ $h \cdot f = W_i + E_k$	$\checkmark$ $\textcircled{a}$
--	-----------------------------------

n	t	x	v	a
0	0.00	2.5	6	-11.01
1	0.05	2.77	5.45	-10.8
2	0.1	3.02	4.81	-10.78
3	0.15	3.24	4.37	-10.68
4	0.2	3.43	3.84	-10.58 m/s ²



② a) $m = 2 \text{ kg}$
 $x = 3.5 \text{ m}$

$$a_0 = \frac{-kv}{m} - g = \frac{-0.16 \cdot 6}{0.8} - 9.81 = -11.01$$

$$v_1 = v_0 + a_0 \cdot \Delta t = 6 - 11.01 \cdot 0.05 = 5.45 \text{ m/s}$$

$$x_1 = x_0 + v_1 \cdot \Delta t = 2.5 + 5.45 \cdot 0.05 = 2.77 \text{ m}$$

$$a_1 = \frac{-kv}{m} - g = \frac{-0.16 \cdot 5.45}{0.8} - 9.81 = -10.78 \text{ m/s}^2$$

$$v_2 = 5.45 - 10.78 \cdot 0.05 = 4.81 \text{ m/s}$$

$$x_2 = 2.77 + 4.81 \cdot 0.05 = 3.02 \text{ m}$$

$$a_2 = \frac{-0.16 \cdot 4.81}{0.8} - 9.81 = -10.78 \text{ m/s}^2$$

$$v_3 = 4.81 - 10.78 \cdot 0.05 = 4.37 \text{ m/s}$$

$$x_3 = 3.02 + 4.37 \cdot 0.05 = 3.24 \text{ m}$$

$$a_3 = \frac{-0.16 \cdot 4.37}{0.8} - 9.81 = -10.68 \text{ m/s}^2$$

$$x_4 = 3.24 + 4.37 \cdot 0.05 = 3.43 \text{ m}$$

$$a_4 = \frac{-0.16 \cdot 3.84}{0.8} - 9.81 = -10.58 \text{ m/s}^2$$

^{2. kol}
 Zadatak 4. Satelit za globalni pozicijski sustav (GPS) obide Zemlju 73 puta mjesečno. → 30 dana ako ne piše

a) Ako gravitacijska sila ima ulogu centripetalne sile izračunajte udaljenost orbite po kojoj se giba taj satelit od središta Zemlje?

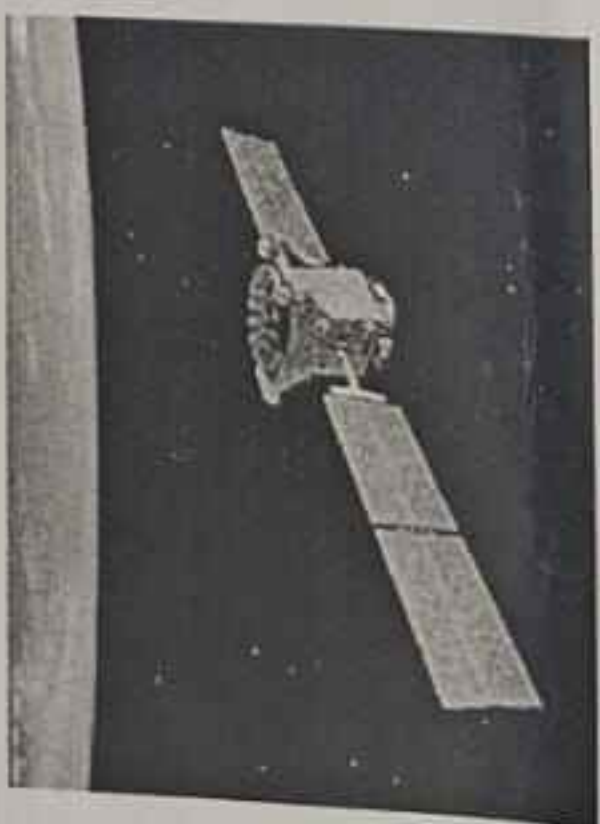
$$m_z = 6 \cdot 10^{24} \text{ kg}$$

$$r_z = 6400 \text{ km}$$

$$R_j. a) 2,3 \cdot 10^7 \text{ m}$$

$$F_g = F_{cp}$$

$$F = \frac{m_z}{30} = 2 \cdot 10^{23} \text{ kg}$$



$$V = \frac{2r\pi}{T} = \frac{2 \cdot 6400 \cdot 10^3 \cdot \pi}{86400} = 2,43 \cdot 10^2 \text{ m/s}$$

$$= 0,12$$

$$V = \frac{2 \cdot 6400 \cdot 10^3 \cdot \pi}{T}$$

$$V = 1500 \text{ m/s}$$

$$F_{cp} = m \cdot a_{cp}$$

$$a_{cp} = \frac{v^2}{r}$$

$$F_g = F_{cp}$$

$$F_g = m \cdot a_{cp}$$

$$8,83 =$$

$$m \cdot a = m \cdot \frac{v^2}{r}$$

$$8,83 = \frac{v^2}{r}$$

$$r = \frac{v^2}{8,83} =$$



Valovi projekcija

$$\frac{v}{m} \rightarrow I$$

2 kol. $\rightarrow$ valovi neće biti žvi!
 Zadatak 5. Kolika je razina jakosti zvuka na udaljenosti 65 metara od aviona ako je razina jakosti zvuka na udaljenosti 85 metara od aviona 105dB?

L - dB

0 - 110 dB

Rj: 107,3dB

$L = ?$

$$L = 10 \log \frac{1}{I_0}$$

$I_1 =$

$r_1 = 65m$

$L_2 = 105dB$

$$I_1 \cdot A_1 = I_2 \cdot A_2$$

$r_2 = 85m$

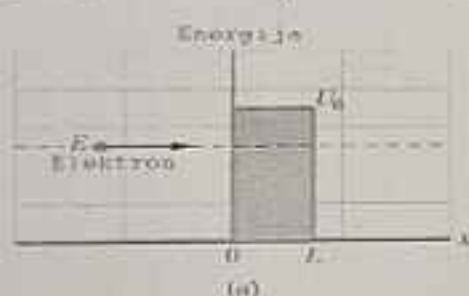
$$A_2 = \left[\frac{4}{3} r^2 \pi \right] \checkmark$$

$$L_2 = 10 \cdot \log_{10} \frac{I_2}{I_0} \rightarrow 10,5 = \log \frac{I_2}{10^{-12}} \cdot 10$$

$$10^{10,5} = \frac{I_2}{10^{-12}} = I_2 = 10^{10,5} \cdot 10^{-12} = 10^{-1,5} \frac{W}{m^2}$$

2. kol.

Zadatak 6. Razmotrite potencijalnu barijeru koju vidite na a) dijelu slike, koja ima visinu $U_0 = 4,2eV$ i debljinu $L = 0,62nm$. Kolika je energija upadnog zračenja elektrona čiji je koeficijent transmisije $T = 0,004$? Energiju izračunajte u eV-ima.



Rj: 3,44eV

$$m_e = 9,11 \cdot 10^{-31} kg$$

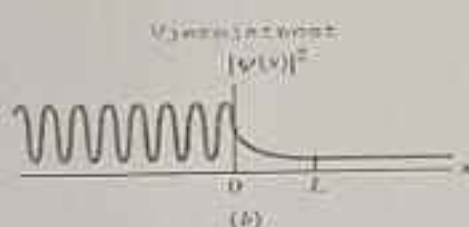
$$1eV = 1,6 \cdot 10^{-19} J$$

$$h = 6,63 \cdot 10^{-34} Js$$

$$\frac{\hbar^2 k^2}{2m} = U_0 - E$$

$$\frac{\hbar^2 k^2}{2m} = U_0 - E$$

$$E = U_0 - \frac{\hbar^2 k^2}{2m}$$



$$U_0 = 4,2eV$$

$$L = 0,62nm = 0,62 \cdot 10^{-9} m$$

$E = ?$

$$T = 0,004$$

$$T = e^{-2\kappa L}$$

$$\kappa = \frac{\ln(T)}{-2L} = 4,45 \cdot 10^9$$

$$\kappa = \sqrt{\frac{2m(U_0 - E)}{\hbar^2}} \rightarrow \kappa^2 = \frac{2m(U_0 - E)}{\hbar^2} \rightarrow \text{izluciti } E$$

2 kol.

Zadatak 7. Metalna kocka gustoće $12 g/cm^3$ uronjena je u vodu gustoće $1 g/cm^3$ i ima prividnu težinu $70 N$. Izračunajte volumen kocke.

$$N = kg/m/s$$

Rj: $0,64 \cdot 10^{-3} m^3$

$V_k = ?$

$$\rho_k = \frac{m_k}{V_k}$$

$$\rho_k = 12 g$$

$$12g = \frac{70N}{V_k}$$

$$\rho_v = 1g/cm^3$$

$$m = 70N$$

$$0,012g = \frac{70}{V_k} \cdot V$$

$$V_k = \frac{70}{0,012}$$

Kakva brzina čam najmanje je od legue gustode 4000 kg/m^3 .
 1. Keži na dnu posude ispunjene vodom. Odredite tlak kojim
 voda djeluje na dno posude. $\rho_{\text{voda}} = 1000 \text{ kg/m}^3$

$$P = \frac{F}{A} = \frac{3.68 \text{ N}}{0.05^2}$$

$$F_g - F_v =$$

$$mg - \rho_{\text{voda}} \cdot V_k \cdot g =$$

$$\rho_k \cdot V_k \cdot g - \rho_{\text{voda}} \cdot V_k \cdot g =$$

$$4000 \cdot 0.05^3 \cdot 9.81 - 1000 \cdot 0.05^3 \cdot 9.81 = 3.68 \text{ N}$$

$$P = \frac{F}{A} = \frac{3.68 \text{ N}}{0.05^2}$$

$$= 1471.5 \text{ Pa}$$

$$a = 5 \text{ cm} = 0.05 \text{ m}$$

$$\rho_k = 4000 \text{ kg/m}^3$$

$$\rho_{\text{voda}} = 1000 \text{ kg}$$

$$F_g = F_v = F_u$$

$$= \rho_k \cdot V_k \cdot g - \rho_v \cdot V_k \cdot g$$

$$= 3.68 \text{ N}$$

$$a = 5 \text{ cm} = 0.05 \text{ m}$$

$$\rho_k = 4000 \text{ kg/m}^3$$

$$\rho_v = 1000 \text{ kg/m}^3$$

$$P = \frac{F}{A}$$

$$F_g - F_v = \rho_k \cdot V_k \cdot g - \rho_v \cdot V_k \cdot g$$

$$= X$$

Kolika je najmanja frekvencija svjetlosti potrebna za fotoelektrični
 efekt ako je $W_i = 3.1 \text{ eV}$? Kolika je valna dužina te svjetlosi

$$E = W_i + E_k$$

$$\lambda = \frac{c}{f} = \frac{3 \cdot 10^8}{6.63 \cdot 10^{-19}}$$

$$h \cdot f = W_i + E_k$$

$$h \cdot f = 3.1 \text{ eV} \cdot 1.6 \cdot 10^{-19} \text{ J}$$

$$= \frac{3 \cdot 10^8}{0.748 \cdot 10^{15} \text{ Hz}}$$

$$6.63 \cdot 10^{-34} \cdot f = 4.96 \cdot 10^{-19} \text{ J}$$

$$= 4.01 \cdot 10^{-7} \text{ m}$$

$$f = 0.748 \cdot 10^{15} \text{ Hz}$$

Kvantna

Kada imamo zraklo U_0 , L , T , a tražimo E formula je:

$$T = e^{2ikL} \rightarrow k = \frac{\ln(1/T)}{-2L} \dots$$

Ona uz pomoć tog k računamo E :

$$k = \sqrt{\frac{8\pi^2 m (U_0 - E)}{h^2}} \rightarrow k^2 = \frac{8\pi^2 m (U_0 - E)}{h^2} \xrightarrow{\cdot h^2}$$

$$\rightarrow k^2 h^2 = 8\pi^2 m (U_0 - E) \xrightarrow{\cdot 1/8\pi^2 m} \frac{k^2 h^2}{8\pi^2 m} = U_0 - E \rightarrow$$

$$\rightarrow E = U_0 - \left(\frac{k^2 h^2}{8\pi^2 m} \right) \rightarrow \text{pretvorit u eV}$$

Kada je zraklo samo U_0 , a traži se ① frekvencija ili ② valna dužina:

$$h \cdot f = W_i$$

$$f = \frac{W_i}{h}$$

$$\rightarrow \lambda = \frac{c}{f}$$

$$c = 3 \cdot 10^8 \text{ m/s}$$

$$t = 20 \text{ days} = 30 \cdot 60 \cdot 24 \cdot 60 = 2.592000 \text{ s}$$

$$T = \frac{2.592000}{43}$$

$$a_{cp} = \frac{v^2}{R}$$

$$v = \frac{2\pi R}{T}$$

$$\begin{aligned} F_g &= F_{cp} \\ F_g &= m \cdot a_{cp} \\ F_g &= m \cdot \frac{v^2}{R} \\ m \cdot a &= m \cdot \frac{v^2}{R} \end{aligned}$$

$$T = \frac{30 \cdot 24 \cdot 60 \cdot 60}{43}$$

$$= 35506.84 \text{ s}$$

$$G \cdot \frac{m_s \cdot M_x}{R^2} = m_s \cdot a_{cp}$$

$$G \cdot \frac{m_s \cdot M_x}{R^2} = m_s \cdot \frac{v^2}{R}$$

$$G \cdot \frac{m_s \cdot M_x}{R^2} = m_s \cdot \frac{4R^2 \pi^2}{T \cdot R}$$

$$G \cdot \frac{m_s \cdot M_x}{R^2} = m_s \cdot \frac{4R \pi^2}{T}$$

$$G \cdot M_x \cdot T = 4r^3 \pi^2 \quad | : 4\pi^2$$

$$r^3 = \frac{G \cdot M_x \cdot T}{4\pi^2}$$

$$\begin{aligned} \cancel{G \cdot m_s \cdot M_x} / R^2 &= \cancel{m_s \cdot M_x} \cdot \frac{4\pi^2 R}{T \cdot \cancel{R}} \\ \frac{G \cdot M_x}{R^2} &= \frac{M_x \cdot 4\pi^2 R}{T} \end{aligned}$$

$$G \cdot \cancel{m_s} \cdot M_x \cdot T = 4R^3 \cdot \cancel{m_s} \cdot \pi^2$$

$$G \cdot M_x \cdot T = 4R^3 \pi^2 \quad | : 4\pi^2$$

$$\sqrt[3]{\frac{G \cdot M_x \cdot T}{4\pi^2}} = R$$

$$R =$$

$$\frac{G \cdot \cancel{m_s} \cdot M_x}{R^2} = \frac{\cancel{m_s} \cdot 4\pi^2 R^2}{T^2 \cdot \cancel{R}} \quad | : m_s$$

$$\frac{G \cdot M_x}{R^2} = \frac{4R^2 \pi^2}{T^2} \quad | \cdot T^2 \rightarrow \frac{G \cdot M_x \cdot T^2}{R^2} = 4R^2 \pi^2 \quad | : 4\pi^2$$

Usgon

$$\rho_k = 12 \text{ g/cm}^3 = 12 \cdot \frac{1000}{10^{-6}} = 12 \cdot \frac{10^6}{10^3} = 12 \cdot 10^3 \text{ kg/m}^3$$

$$\rho_{\text{voda}} = 1 \text{ g/cm}^3 = 1000 \text{ kg/m}^3$$

$$F_g' = 70 \text{ N}$$

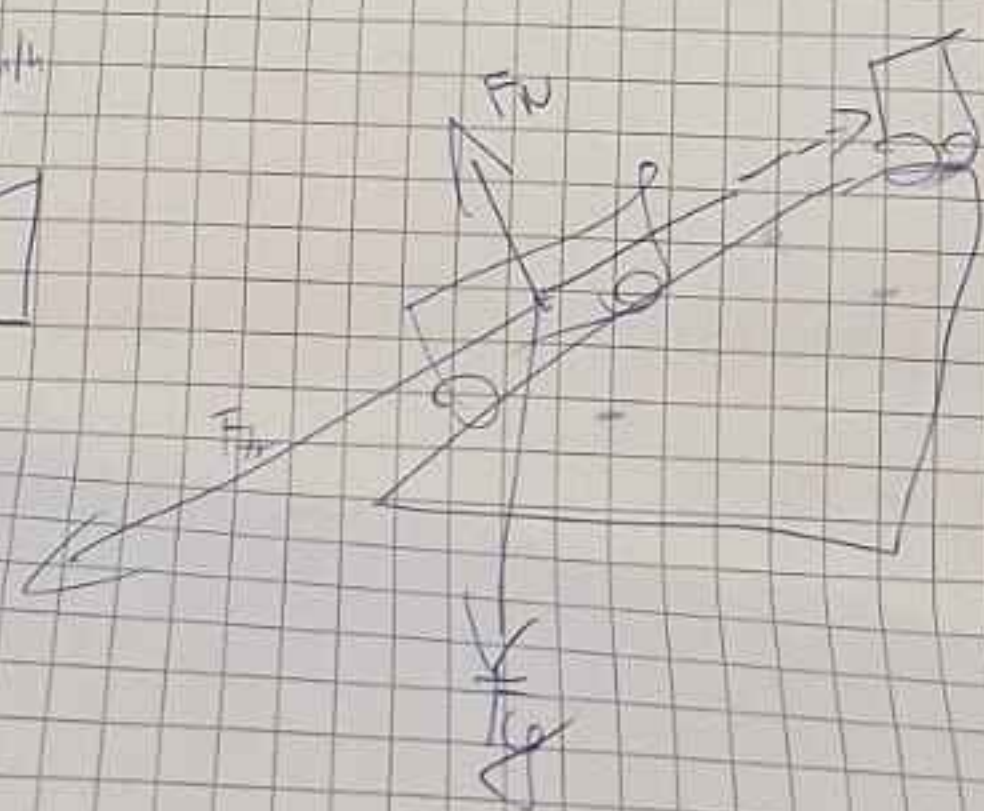
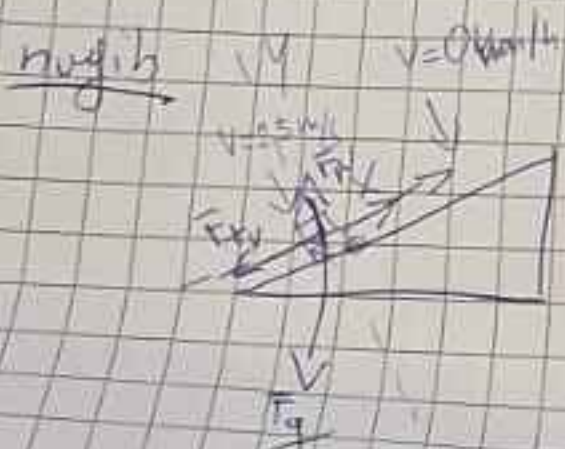
$$V_k = ?$$

$$F_g' = F_g - F_{\text{vzgon}}$$

$$70 = \rho_k \cdot V_k \cdot g - \rho_{\text{voda}} \cdot V_k \cdot g$$

$$70 = V_k \cdot g (\rho_k - \rho_{\text{voda}})$$

$$V_k = \frac{70}{g (\rho_k - \rho_{\text{voda}})} = 0.64 \cdot 10^{-3} \text{ m}^3$$



$$\sin 15 = \frac{F_{tr}}{F_g}$$

$$F = m \cdot a$$

$$-F_{tr} - F_{gx} = m \cdot a$$

$$\cos 15 = \frac{F_{gx}}{F_g}$$

$$- \mu \cdot m \cdot g \cdot \cos 15 - m \cdot g \cdot \sin 15 = m \cdot a$$

$$a = -3.48 \text{ m/s}^2$$

$$v^2 = v_0^2 + 2ax$$

$$2ax = v_0^2$$

$$-6.97x = 90 \text{ km/h} = 25 \text{ m/s}$$

$$x = \frac{25^2}{-6.97} \approx -89.6$$

gravitacija:

$$\vec{F}_g = \vec{F}_{cp} \rightarrow \vec{F}_{cp} = m \cdot a_{cp} \rightarrow a_{cp} = \frac{a_v^2}{R} \rightarrow v = \frac{2\pi R}{T}$$

$$\vec{F}_g = m \cdot \frac{4\pi^2 R}{T^2}$$

R je ekv. razdalje r od satelita do površine Zemlje

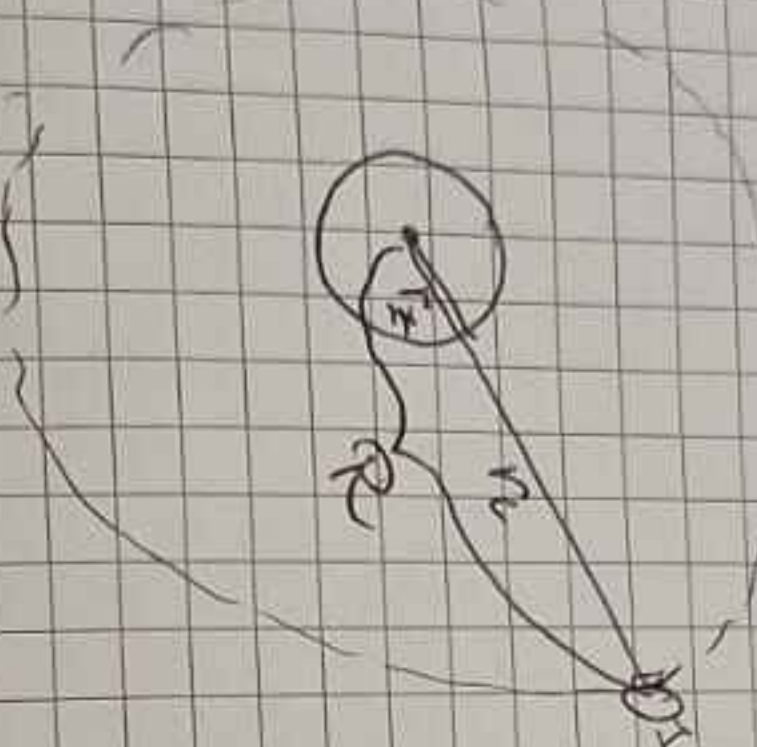
$\vec{F}_g$ je sila tek. konstante

$$\vec{F}_g = G \cdot \frac{m_s \cdot M_z}{R^2}$$

Končni zapis je

$$G \cdot \frac{m_s \cdot M_z}{R^2} = m_s \cdot \frac{4\pi^2 R}{T^2}$$

preko te formule dobijemo R



ako imamo zadano koliko puta objekt obije x sek dana
ili neko drugo vrijeme:

$$T = \frac{y \rightarrow v \text{ sekundama}}{\text{puta}} \rightarrow T = x \text{ sekundi}$$

$$\begin{aligned} \textcircled{2} \quad r_1 &= 65 \text{ m} \\ L_1 &= ? \\ r_2 &= 85 \text{ m} \\ L_2 &= 105 \text{ dB} \\ \Delta L &= 0.3 \end{aligned}$$

$$I_1 \cdot A_1 = I_2 \cdot A_2$$

$$A = \frac{4}{3} r^2 \pi$$

$$\begin{aligned} L_2 &= 10 \cdot \log \frac{I_2}{I_0} = \\ &= 10.5 = \log \frac{I_2}{10^{-12}} = \\ &= 10^{10.5} = \frac{I_2}{10^{-12}} \end{aligned}$$

$$I_2 = 10^{10.5} \cdot 10^{-12} = 10^{-1.5} \text{ W/m}^2$$

$$I_1 \cdot \frac{4}{3} \cdot 65^2 \cdot \pi = 10^{-1.5} \cdot \frac{4}{3} \cdot 85^2 \cdot \pi$$

$$I_1 \cdot 17697.6 = 10^{-1.5} \cdot 30264.01$$

$$I_1 = \frac{10^{-1.5} \cdot 30264.01}{17697.6}$$

$$I_1 = 0.054 \text{ W/m}^2$$

$$L_1 = 10 \cdot \log \frac{I_1}{I_0}$$

$$L_1 = 10 \cdot \log \frac{0.054}{10^{-12}} = 32 \text{ dB}$$

$$T = e^{-2\kappa L} / \ln$$

$$\kappa = \frac{\ln(T)}{-2L}$$

$$\kappa = 4.45 \cdot 10^9$$

$$\textcircled{3} \quad U_0 = 9.2 \text{ eV}$$

$$L = 0.62 \text{ nm} = 0.62 \cdot 10^{-9} \text{ m}$$

$$T = 0.004$$

$$\kappa = \sqrt{\frac{8\pi^2 m (U_0 - E)}{h^2}} \rightarrow \kappa^2 = \frac{8\pi^2 m (U_0 - E)}{h^2} = \frac{h^2}{8\pi^2 m} (U_0 - E)$$

$$= \frac{h^2}{8\pi^2 m} (U_0 - E) \rightarrow E = U_0 - \frac{h^2}{8\pi^2 m}$$

$$E = 9.2 - 0.76 = 3.94 \text{ eV}$$

Valori period

$$V = \sqrt{\frac{F}{\rho A}}$$

$$A = \pi r^2$$

$$n = \frac{V}{f}$$

$$f = \frac{w}{2\pi}$$

$$w = \frac{\Delta \phi}{\Delta t}$$

$$\Delta \phi = \alpha \cdot \frac{\pi}{180} \text{ rad}$$

$$x_n = (n-1) \cdot \frac{\pi}{2}$$

$$\Delta L = L_2 - L_1$$

Kada imamo ΔL , nekā L i nekā r formula je:

$$L = 20 \cdot \log\left(\frac{r_2}{r_1}\right) \rightarrow 20 \text{ jer prebacujemo } \frac{I}{I_0} \text{ u } \frac{r_2}{r_1}$$

Kada imamo r_1, r_2 i L_2 formula je:

$$L_2 = 10 \cdot \log \frac{I_2}{I_0} \xrightarrow{/:10} \frac{L_2}{10} = \log \frac{I_2}{I_0} \xrightarrow{:\log} 10^{\frac{L_2}{10}} = \frac{I_2}{I_0} \dots$$

$$I_0 = 10^{-12} \text{ W/m}^2$$

$$I_1 \cdot A_1 = I_2 \cdot A_2 \rightarrow A_1 = \frac{4}{3} r_1^2 \pi, \quad A_2 = \frac{4}{3} r_2^2 \pi$$

↓ ↓
priredit na r

④ At

$$h\nu = 100 \text{ m}$$

$$h\nu = ?$$

$$\nu = \frac{c}{\lambda}$$

$$\sqrt{2gzh} = \sqrt{2gzh_{\text{max}}}$$

$$2gzh = 2gzh_{\text{max}}$$

$$2gzh = 2 \cdot \frac{1}{6} g z h_{\text{max}}$$

$$h_{\text{max}} = 6h_z = 600 \text{ m}$$

⑤

$$I_0 = 10 \text{ mW/cm}^2$$

$$I_2 = 500 \text{ mW/cm}^2$$

$$\Delta I = 10 \log \frac{I_2}{I_1} - 10 \log \frac{I_1}{I_0}$$

$$= 10 \log \frac{I_2}{I_1} = 16.99 \text{ dB}$$

V

⑥

$$W_i = 4 \text{ eV}$$

$$V_{\text{max}} = 6 \cdot 10^6 \text{ m/s}$$

$$h = ?$$

$$h\nu = W_i + E_{k \text{ max}}$$

$$h\nu = W_i + E_{k \text{ max}}$$

$$\frac{hc}{\lambda} = W_i + E_{k \text{ max}}$$

$$E = h \cdot \nu$$

$$E_{k \text{ max}} = 102.49 + 4 = 106.49 \text{ eV}$$

$$\nu = \frac{E}{h} = 2.57 \cdot 10^{16} \text{ s}^{-1}$$

$$E_k = \frac{mv^2}{2} = 28.26 \cdot 10^{-36} \text{ eV}$$

$$= 102.49 \text{ eV}$$

a) $W_i = ?$

$\lambda = 125 \text{ nm}$

$E_{\text{ke}} = 0.4 \text{ eV}$

$h \cdot f = W_i + E_{\text{kin}}$

$W_i = \frac{hc}{\lambda} - E_{\text{ke}}$

$= 1.08 \cdot 10^{-18} - 0.4 \text{ eV}$

$= 6.32 \text{ eV}$

b)

$U = 0.40 \text{ eV}$

E

$T = 0.02$

$E = \frac{mv^2}{2}$

$2E = mv^2$

$v^2 = \frac{2E}{m} = 1.41 \cdot 10^{11}$

$= 3.75839 \cdot 10^5 \text{ m/s}$

$k = \sqrt{\frac{8\pi^2 m (U_0 - E)}{h^2}} = 2894513015$

$T = e^{-2kL}$

$L = \frac{\ln T}{-2k} = 0.68 \text{ nm} = 6.754653162 \cdot 10^{-10} \text{ m}$

c)

$I_2 = 129 \text{ dB}$

$r_2 = ?$

$r_1 = 97 \text{ m}$

$I_1 = 119.44 \text{ dB}$

$I_1 \cdot r_1^2 = I_2 \cdot r_2^2$

$119.44 \cdot 97^2 = 129 \cdot r_2^2$

$1123810.96 = 129 r_2^2$

$= 8711.71$

$r_2 = 93.39 \text{ m}$

$P = 1 \text{ A}$

$= 129 \cdot 4 \cdot 93.39^2 \cdot \pi$

=

Novi r preko $I_1 \cdot r_1^2 = I_2 \cdot r_2^2$

1. taj dobiveni r koristi u $P = I \cdot r$

2. e od ovog dobijemo (129 dB)

Primer ispit

①

n	t	x	v	a
0	0	2.5	6	-11.0
1	0.05	2.77	5.45	-10.9
2	0.1	3.02	4.91	-10.8
3	0.15	3.28	4.37	-10.7
4	0.2	3.55	3.83	-10.6

$$a_0 = \frac{-kv}{m} - g = -11.03 \text{ m/s}^2$$

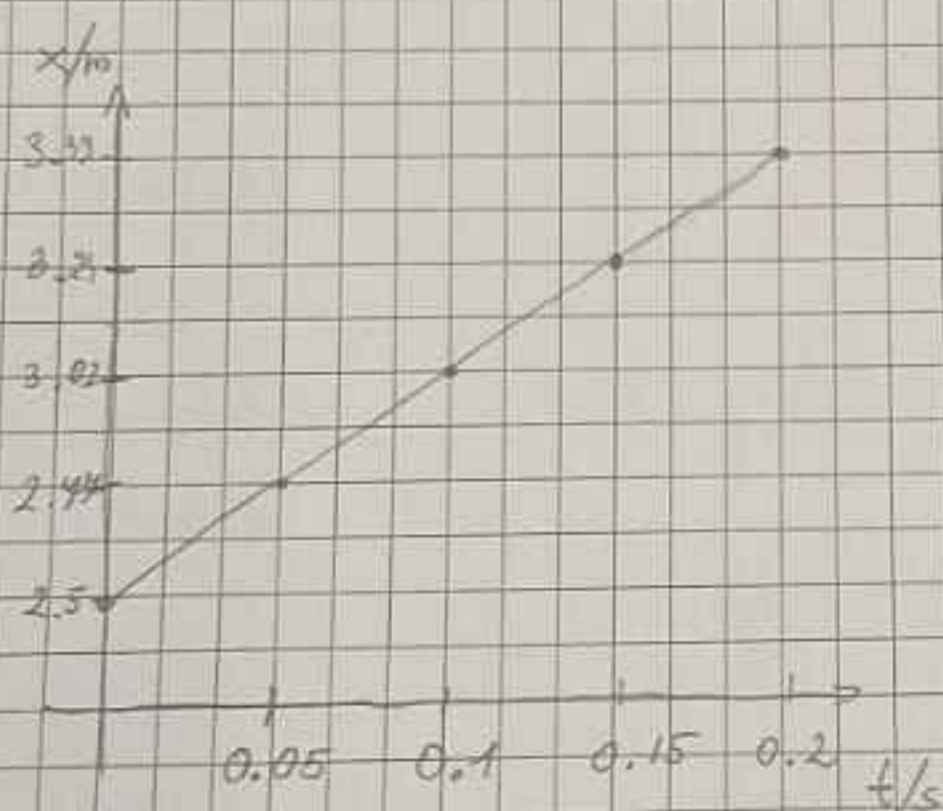
$$v_1 = v_0 + a_0 \cdot \Delta t = 5.45 \text{ m/s}$$

$$x_1 = x_0 + v_1 \cdot \Delta t = 2.77 \text{ m}$$

$$a_1 = \frac{-kv_1}{m} - g = -10.9 \text{ m/s}^2$$

$$v_2 = v_1 + a_1 \cdot \Delta t = 4.91 \text{ m/s}$$

$$x_2 = x_1 + v_2 \cdot \Delta t = 3.02 \text{ m}$$



②

a) $m = 2 \text{ kg}$

$F = 30.5 \text{ N}$

$\Delta_1 = 6$

$14 + 10 + 6.5$

$x = 3.5$

$\Delta_2 = 11$

$20.5 + 10 = 30.5$

$\Delta_3 = 10$

$\Delta_4 = 6.5$

b) $E_k = \frac{mv^2}{2} = \frac{2 \cdot 4^2}{2} = 16 \text{ J}$

$v = 4 \text{ m/s}$

$E_v = 16 + 30.5$

$= 46.5 \text{ J}$

5) a) $A = 1.253 \text{ cm/s}^2$ $a = 2 \cdot A$
 $f_n = 30$ $a = \frac{2A}{f_n^2} = 12.7 \text{ m/s}^2$
 $f_k = 650$

b) $m = 2.45 \text{ kg}$
 $F = m \cdot a$
 $= 2.45 \cdot 12.7$
 $= 31.12 \text{ N}$

c) $I = F \cdot \Delta t$
 $= 31.12 \cdot \frac{0.08}{\frac{30}{650}} = 53.94 \text{ Ns}$

d) $v_1 = \frac{\Delta x}{\Delta t} = 18.08 \text{ m/s}$
 $v_2 = \frac{\Delta x}{\Delta t} = 21.1 \text{ m/s}$

6) $\rho = 3950 \text{ kg/m}^3$ $g_z = G \cdot \frac{M_z}{r_z^2} = 9.774 \text{ m/s}^2$
 $D = 6730 \text{ km}$ $g_M = G \cdot \frac{M_M}{r_M^2} = 3.94$
 $h = 10 \text{ m}$

$$\rho = \frac{m}{V}$$

$$m = \rho \cdot V$$

$$= 3950 \cdot \frac{4}{3} \cdot 3395^3 \pi = 6.47 \cdot 10^{13}$$

$$= 3950 \text{ kg/m}^3 \cdot \frac{4}{3} \cdot (3395 \text{ m})^3 \cdot \pi = 6.47 \cdot 10^{13} \text{ kg}$$

$$\frac{g_M}{g_z} = 0.38$$

$$v = \sqrt{2gh}$$

$$v = 8.65 \text{ m/s}$$

5

$$I_1 = 115 \text{ dB} \quad r_1 = 85$$

$$I_2 = 100 \text{ dB} \quad r_2 = ?$$

$$I_0 = 10^{-12} \text{ W/m}^2$$

$$I_1 = 10^{11.5 - 12} = 10^{-0.5} \text{ W/m}^2$$

$$I_2 = 10^0 - 10^{-12} = 10^{-2} \text{ W/m}^2$$

$$I_1 \cdot r_1^2 = I_2 \cdot r_2^2$$

$$10^{-0.5} \cdot 85^2 = 10^{-2} \cdot r_2^2$$

$$2289.44$$

$$r_2 = 448 \text{ m}$$

$$P = 1 \text{ A}$$

$$P = 10^{-0.5} \cdot 4\pi r_2^2$$

$$= 28410.96 \text{ W}$$

6

$$U_0 = 6 \text{ eV}$$

$$L = ?$$

$$E_{02} = 5.2 \text{ eV}$$

$$T = 0.001$$

$$K = \sqrt{\frac{8\pi^2 m (U - E)}{h^2}}$$

$$= \sqrt{\quad} = 4576626.323$$

$$L = \frac{h(T)}{2K} = 6.48 \cdot 10^{-10} \text{ m}$$

$$= 0.68 \text{ nm}$$

① $h = 1200\text{m}$

$d = 500\text{m}$

$$t^2 = \frac{2H}{g} = \frac{2400}{9.81} = 244.65 \text{ s}^2$$

$$= 15.64\text{s}$$

$x = v_0 t$

$v_0 = \frac{x}{t} = \frac{500}{15.64} = 31.97\text{m/s}$

② $h = 50\text{m}$

$v_0 = 30\text{m/s}$

a) $T = \sqrt{\frac{2H}{g}} = \sqrt{\frac{100}{9.81}} = 3.19\text{s}$

$D = v_0 t$

$D = 30 \cdot 3.19 = 95.78\text{m}$

③ $h = 30\text{m}$

$v_0 = 10\text{m/s}$

$T = \sqrt{\frac{2H}{g}} = \sqrt{\frac{60}{9.81}} = 2.42\text{s}$

$D = v_0 \cdot t$

$= 24.2\text{m}$

④

n	t	x	v	a
0	0	1.8	12	-17.01
1	0.05	2.36	11.15	-16.03
2	0.1	2.88	10.35	-15.14
3	0.15	3.36	9.59	-14.41
4	0.2	3.80	8.84	-13.74

$$⑤ \quad r_u = 4850 \text{ m}$$

$$m = 0.0558 \cdot m_x = 3.35 \cdot 10^{23} \text{ kg}$$

$$h_u = ?$$

$$h_x = 10 \text{ m}$$

$$a_u = G \frac{M}{r^2} = 0.44 \text{ m/s}^2$$

$$m_u g_u h_u = m_m g_m h_m$$

$$h_m = \frac{g_u h_u}{g_m}$$

$$h_m = 109.36 \text{ m}$$

$$⑥ \quad \beta_2 = 128 \text{ dB}$$

$$\beta_1 = 1.08 \cdot \beta_2 = 138.32 \text{ dB}$$

?

$$⑦ \quad m_i = 300 \text{ kg}$$

$$a_i = G \cdot \frac{1.35 \cdot 10^{23}}{2545^2}$$

$$= 1.41 \text{ m/s}^2$$

$$\frac{a_i}{a_{ze}} = 0.14$$

$$F = m \cdot a$$

$$a_{\#n} = G \cdot \frac{M}{r^2} = 9.86$$

$$= 423 \text{ N}$$

$$a_{ze} = 9.84 \text{ m/s}^2$$

⑧

$$W_i = 3.1 \text{ eV}$$

$$v = \frac{W_i}{h}$$

$$r = ?$$

$$= 7.49 \cdot 10^{14} \text{ s}$$

$$\lambda = ?$$

$$\lambda = \frac{c}{v}$$

$$= \frac{3 \cdot 10^8}{7.49 \cdot 10^{14}} = 4.005 \cdot 10^{-7} \text{ m}$$

Kvantna mehanika

$$L = \frac{h \cdot f}{2\pi}$$

$$\lambda = \frac{c}{\nu}$$

$$\gamma = \frac{W_i + E_{\text{uv}}}{h}$$

$$W_i = \frac{\Delta p}{\Delta t} = \frac{p}{\Delta t}$$

$$\lambda = \frac{v}{f}$$

$$f = \frac{W_i}{2\pi}$$

$$h\nu = W_i + E_{\text{uv}}$$

$$v = \lambda \cdot f$$

$$T = \frac{1}{f}$$

$$W_i = \frac{h\nu}{\lambda} + E_{\text{uv}}$$

$$v = \frac{E}{h}$$

$$E = h \cdot \nu$$

$$\frac{h\nu}{\lambda} = W_i + E_{\text{uv}}$$

Šum

$$\Delta L = 10 \log \frac{I_2}{I_0} - 10 \log \frac{I_1}{I_0}$$

$$= 10 \log \frac{I_2}{I_1}$$

Opći zakon gravitacije

$$v = \sqrt{2gh} \Leftrightarrow v = \sqrt{2gh}$$

$$2g_1 h_1 = 2g_2 h_2$$

$$h_1 = \frac{g_2}{g_1} \cdot h_2$$

Faktor reprodukcije

$\frac{f_r}{f_s} = f_k$ } faktor korekcije je nu kvadrat samo pri istom akceleraciji
U svim drugim slučajevima je bez potencije

Ispit 8. rok

① $m_{kolica} = 20 \text{ kg}$ $m_{dj} = 40 \text{ kg}$
 $v_{kolica} = 1 \text{ m/s}$ $v_{dj} = 2 \text{ m/s}$

$$m_k \cdot v_k - m_d \cdot v_{dj} = m_k \cdot v_{nk} + m_d \cdot v_{ndj}$$

$$20 \cdot 1 - 40 \cdot 2 = 20 \cdot v + 40 \cdot 0$$

$$-60 = 20v$$

$$v = -3 \text{ m/s}$$

Orientacija kolica i djevojčice je obrnuta početnoj

② $f_r = 25$ $m_c = 0.01 \text{ kg}$

$$f_s = 500$$

a) $\lambda_{2-5} = 982 - 267 = 715 \text{ mm} = 0.715 \text{ m}$

b) $A = 24.035 \text{ mm/s}^2$

$$a = 2A = \frac{2 \cdot 24.035}{(f_k)^2} = 18246 \text{ mm/s}^2 = 18.28 \text{ m/s}^2$$

c) $v_1 = \frac{\Delta x}{\Delta t} = \frac{967 - 109}{0.1} = 31.6 \text{ m/s}$

$$v_2 = \frac{\Delta x}{\Delta t} = \frac{0}{0.1} = 0 \text{ m/s}$$

$$v_{n-1} = \frac{\Delta x}{\Delta t} = \frac{982 - 715}{0.1} = 46.2 \text{ m/s}$$

d) $E_{k1} = \frac{mv^2}{2}$
 $= \frac{0.01 \cdot 31.6^2}{2} = 4.993 \text{ J}$

$$E_{k2} = \frac{mv^2}{2} = \frac{0.01 \cdot 46.2^2}{2} = 10.672 \text{ J}$$

$$E_p = \frac{m \cdot v^2}{2} = \frac{0.01 \cdot 0}{2} = 0 \text{ J}$$

③

$$m = 1 \text{ kg}$$

$$F = 10.81 \text{ N}$$

$$h = ?$$

$$t = 10 \text{ s}$$

$$g = 9.81 \text{ m/s}^2$$

$$F = m \cdot a$$

$$a = \frac{F}{m}$$

$$a = 10.81 \text{ m/s}^2$$

$$s = \frac{1}{2} a t^2$$

$$s = \frac{1}{2} \cdot 10.81 \cdot 100$$

$$s = 540.5 \text{ m}$$

TO DO

Formula za izračun
visine koje će tijelo
pod stalnom silom imati nakon
 n sekundi:

$$s = \frac{1}{2} \left(\frac{F}{m} - g \right) \cdot t^2$$

$$s = \frac{1}{2} \cdot \left(\frac{10.81}{1} - 9.81 \right) \cdot 100$$

$$= 50 \text{ m}$$

④

$$V = 1000 \text{ cm}^3 = 0.001 \text{ m}^3$$

$$F = 10 \text{ N}$$

$$g = 9.81 \text{ m/s}^2$$

$$\rho = 10^3 \text{ kg/m}^3$$

Arhimedov zakon

$$F_b = \rho \cdot V \cdot g$$

$$= 1000 \text{ kg/m}^3 \cdot 0.001 \text{ m}^3 \cdot 9.81 \text{ m/s}^2$$

$$F_{\text{uzgon}} = 9.81 \text{ N}$$

Sila dinamometra

$$F_d = F_{\text{tijelo}} - F_{\text{uzgon}} = 10 - 9.81 = 0.19 \text{ N}$$

- Uzgon se javlja zbog razlike u pritisku na različitim dijelovima tijela uronjenog u tekućinu

⑤

$$m_1 = 100 \text{ kg}$$

$$m_2 = 50 \text{ kg}$$

a) Crtež B Svaka sila ima jednaku protusilu

$$b) F = \frac{m_1 \cdot m_2}{r^2} \cdot G \quad F = 6.67 \cdot 10^{-11} \cdot \frac{100 \cdot 50}{1000}$$

$$= 3.335 \cdot 10^{-13} \text{ N}$$

$$\textcircled{5} \quad W_i = 4 \text{ eV}$$

$$v_{\text{max}} = 6 \cdot 10^6 \text{ m/s}$$

$$E_k = \frac{1}{2} \cdot m_e \cdot v_e^2$$

$$= \frac{1}{2} \cdot 9.1 \cdot 10^{-31} \cdot (6 \cdot 10^6)^2$$

$$= 1.64 \cdot 10^{-17} \text{ J} = 102.5 \text{ eV}$$

$$f = \frac{W_i + E_k}{h} = 1.61 \cdot 10^{15} \text{ Hz}$$

$$E_k = h \cdot f$$

$$E_k = 106.4 \text{ eV}$$

$$E_k = \frac{1}{2} m \cdot v^2$$

$$v = \sqrt{\frac{2E}{m}}$$

$$v = 6123213.465 \text{ m/s}$$

$$hf = 102.5 + 4 = 106.4 \text{ eV}$$

Energija fotona je proporcionalna frekvenciji

$$hf' = 2 \cdot 106.4 \text{ eV} = 212.8 \text{ eV}$$

$$E_k' = hf' - W_i$$

$$= 212.8 - 4 = 208.8 \text{ eV}$$

$$E_k' = \frac{1}{2} m v^2$$

$$v = \sqrt{\frac{2E}{m}} = 8568780.803$$

$$= 8.56 \cdot 10^6 \text{ m/s} //$$